

PROJECT REPORT

ON

LEAF & BLOOM

An Online Plant Store with Community, Care Guidance, and Plant Assistant

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

Name: _____

Roll No: _____

Under the Guidance of

Guide Name: _____

Department of Computer Science and Engineering

College Name: _____

Academic Year: 2025 - 2026

CERTIFICATE

This is to certify that the project report entitled "**Leaf & Bloom**" is a bonafide work carried out by _____ bearing Roll No. _____ in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work embodied in this report has been carried out under my supervision during the academic year 2025 - 2026.

Project Guide

Head of Department

Internal Examiner

External Examiner

DECLARATION

I hereby declare that the project report entitled "**Leaf & Bloom**" submitted by me to the Department of Computer Science and Engineering is an original work carried out by me under the guidance of _____. This report has not been submitted earlier to any other institution or university for the award of any degree or diploma.

Signature of the Student

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, faculty members, and the Department of Computer Science and Engineering for their continuous support, motivation, and encouragement throughout the development of this project. I also thank my friends and family for their valuable suggestions and moral support during every phase of the work.

This project helped me strengthen both my technical understanding and my confidence in building a complete web application that combines e-commerce, community interaction, and practical plant care assistance.

ABSTRACT

Leaf & Bloom is a modern web-based plant shopping platform designed to make plant discovery, care guidance, and community participation simple and engaging. Traditional online shopping platforms often focus only on product listing and order placement, but plant buyers usually need more support, especially beginners who may not know how to care for different plant types. This project addresses that gap by combining a plant catalog, personalized plant guidance, a beginner care guide, wishlist management, order handling, user authentication, and a community space where plant lovers can share their journey and experiences.

The frontend of the system is developed using React and Vite to provide a fast and responsive user experience, while the backend uses Node.js, Express.js, and MongoDB to manage order data and wishlist-related information. The system includes category browsing, room-based discovery, plant detail pages, reviews, chatbot-style assistance, community posts with images, and structured plant care information. Leaf & Bloom is designed not only as an e-commerce solution but also as a friendly digital companion for plant lovers.

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1. INTRODUCTION

Indoor plants have become an important part of modern lifestyles because they improve aesthetics, support well-being, and create a closer connection with nature. However, many buyers struggle to choose the right plant, understand its care requirements, and maintain it successfully after purchase. General e-commerce websites usually do not offer detailed plant-specific guidance or community-based support.

Leaf & Bloom is developed as a specialized online plant store that offers a complete experience around plant shopping and care. It helps users browse plants based on category, room suitability, lighting conditions, beginner friendliness, and lifestyle preferences. In addition to shopping, the platform provides care guidance, a community space, and a context-aware plant assistant to make the experience more helpful and personalized.

2. PROBLEM STATEMENT

Existing shopping platforms treat plants like ordinary products and often fail to support customers with practical plant-care knowledge. Users may purchase plants without understanding light needs, watering frequency, humidity, pet safety, or maintenance difficulty. This leads to poor plant survival, customer dissatisfaction, and a low level of confidence among beginners. There is a need for a dedicated digital platform that combines plant shopping with educational guidance and user interaction.

3. OBJECTIVES

- To design a dedicated online platform for buying indoor and lifestyle plants.
- To provide structured plant-specific guidance for beginners and regular plant owners.
- To implement shopping features such as catalog browsing, wishlist, checkout, and order history.
- To add a plant community where users can share experiences, images, and joy.
- To offer a plant assistant that responds to care questions using plant-specific context.
- To build an interface that is visually attractive, simple to navigate, and mobile friendly.

4. EXISTING SYSTEM

In the existing system, users normally rely on generic e-commerce platforms or scattered social media pages to buy plants. Product information is often limited to names, prices, and a few images. Plant care support is inconsistent, community interaction is disconnected from the buying process, and beginner guidance is rarely integrated into the shopping flow. As a result, buyers often feel uncertain before and after purchase.

5. PROPOSED SYSTEM

The proposed system, Leaf & Bloom, integrates plant shopping and support into one platform. It includes a rich product catalog, category-based navigation, room-based browsing, reviews, wishlist management, checkout, and order management. It also includes a care guide, a plant assistant for customized suggestions, and a community page for shared learning and engagement. This makes the platform more useful than a standard storefront because it supports both plant selection and plant success after purchase.

6. SYSTEM REQUIREMENTS

6.1 Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 20 GB free space
- Internet connection for browser access and media loading

6.2 Software Requirements

- Operating System: Windows 10/11
- Frontend: React, Vite, HTML5, CSS3, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB
- Code Editor: Visual Studio Code
- Browser: Google Chrome / Microsoft Edge

7. SYSTEM DESIGN AND ARCHITECTURE

The system follows a client-server architecture. The React frontend handles user interaction, rendering, filtering, and browsing experiences. The Express backend handles API requests and persists order and wishlist data in MongoDB. Static and structured plant data is used to enrich the user experience with care guidance and category browsing.

Layer	Description
Presentation Layer	React-based UI for catalog, plant details, login, wishlist, checkout, care guide, and community.
Application Layer	Business logic for filtering, sorting, review handling, assistant responses, and user interaction flows.
API Layer	Express routes for orders and wishlist management.

Data Layer	MongoDB collections for order data and wishlist items, plus structured plant data for product information.
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8. MODULES DESCRIPTION

8.1 Authentication Module

This module provides Sign In and Sign Up pages for user entry into the system. It creates the basic onboarding flow needed for a personalized shopping experience.

8.2 Plant Catalog Module

This module displays all plant products with filters such as size, light requirement, price range, difficulty, and category. It supports search, sorting, plant collection counters, and category shortcuts such as low light, palms, pet friendly, flowering plants, gift sets, succulents, and trailing plants.

8.3 Plant Details Module

Each plant has a dedicated page that presents detailed care information including watering, humidity, pet safety, placement, soil type, temperature, and fertilizer needs. This module also supports related browsing and plant reviews.

8.4 Wishlist Module

Users can save plants they are interested in and revisit them later. This helps improve decision making and encourages return visits to the platform.

8.5 Checkout and Orders Module

This module handles purchasing flow and stores order information through the backend. It allows users to place orders and view their order history.

8.6 Care Guide Module

This module gives beginner-friendly advice on plant maintenance, including watering routines, light conditions, common issues, and recovery tips for plant stress.

8.7 Plant Assistant Module

A chatbot-style assistant helps users understand how to care for specific plants. It responds with customized guidance based on the active plant and structured plant data.

8.8 Community Module

Users can share their plant journeys, joy, experiences, and images through a community feed. This improves engagement and creates a sense of belonging for plant lovers.

9. DATABASE DESIGN

MongoDB is used as the backend database for order placement and wishlist information. The data model is simple, flexible, and suitable for storing product-related user actions.

Collection	Sample Fields
orders	customerName, phone, address, plantId, plantName, quantity, totalAmount, createdAt
wishlist-items	plantId, name, image, price, category, light, size

Additional structured plant data is maintained in the frontend to support care details, plant categories, and assistant responses.

10. TESTING AND VALIDATION

The application was tested by running the frontend build, checking route rendering, validating UI flows, and verifying backend connectivity. Product listing, search, filtering, wishlist behavior, assistant interaction, plant details, and community posting were reviewed to ensure consistent user experience.

Test Case	Expected Result	Status
Open catalog	Plant listing loads with categories and filters	Pass
Add plant to wishlist	Selected plant appears in wishlist	Pass
Open plant detail page	Plant-specific data and care guidance are shown	Pass
Place order	Order data is stored and visible in orders page	Pass
Use plant assistant	Context-aware care guidance is returned	Pass
Create community post	Post with text and image appears in community feed	Pass

11. ADVANTAGES OF THE SYSTEM

- Dedicated experience for plant shopping instead of generic e-commerce behavior.
- Helpful for beginners through guided care content and plant assistant responses.
- Improves user engagement with community interaction and reviews.
- Supports easier product discovery through categories, room-based browsing, and filters.
- Provides structured plant information that increases customer confidence.

12. APPLICATIONS

- Online indoor plant shopping platforms
- Nursery digital storefronts
- Plant education and beginner care portals
- Community platforms for hobby growers and home gardeners

13. FUTURE SCOPE

- Admin dashboard for adding and editing plant inventory dynamically
- Real user authentication and profile management
- Online payment gateway integration
- Shared backend-powered community feed and reviews
- AI-powered disease diagnosis from uploaded plant images
- Order tracking and delivery notifications

14. CONCLUSION

Leaf & Bloom is a complete and user-friendly web application that brings together shopping, guidance, and community for plant lovers. It solves the limitations of generic plant selling websites by offering meaningful support before and after purchase. The project demonstrates how a focused digital experience can improve customer confidence, simplify plant care for beginners, and create a stronger connection between users and the products they buy.

The system successfully combines modern frontend development, backend integration, structured product data, and engaging user features. It can serve as a strong foundation for future expansion into a more advanced and scalable plant commerce platform.

15. REFERENCES

- React Documentation - <https://react.dev/>
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